

Private Neighbors in Grids and Trees

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Abstract

We study private neighbor parameters, introduced by Hedetniemi and Rall [arXiv:2511.07248], which maximize the number of vertices that are external, internal, or self private neighbors with respect to a set U . These parameters provide a common framework connecting domination, irredundance, independence, and related extremal invariants. In their work, Hedetniemi and Rall posed a problem asking for the best possible constant c such that $ESPN(T) \geq c \cdot n(T)$ for every tree T , and they conjectured exact values for several grid families. We prove that the optimal tree constant is $c = 4/5$ and construct an infinite family of trees showing that this bound is sharp. We also confirm their grid conjectures by proving that $EIPN(P_2 \square P_m) = 2m$, $EISPN(P_3 \square P_{2k}) = 6k$, and $EISPN(P_4 \square P_m) = 4m$. Our proofs combine a structural induction for trees with explicit periodic constructions for grid graphs, giving complete answers to the stated problem and conjectures.

Keywords: private neighbors; irredundance; trees; grid graphs; domination

Mathematics Subject Classifications: 05C69, 05C05

1 Introduction

Domination theory is one of the central themes in graph theory. If G is a graph and $D \subseteq V(G)$, then D is a *dominating set* if every vertex outside D has a neighbor in D . The classical domination number $\gamma(G)$ and its many variants have been developed over several decades; see the monograph of Haynes, Hedetniemi and Henning [12] for a recent systematic treatment. A recurring idea in this area is that minimality of a dominating or irredundant set is witnessed locally by private neighbors. This notion goes back to Cockayne, Hedetniemi and Miller [6], who introduced private neighbors in their study of hereditary hypergraphs and middle graphs.

Let $U \subseteq V(G)$. A vertex x with $p_U(x) = 1$ is called a *private neighbor* with respect to U . It is *external* if $x \notin U$ and *internal* if $x \in U$. Following Hedetniemi and Rall [13], we also count the special case in which $x \in U$ and $p_U(x) = 0$; such a vertex is called a *self private neighbor*. These three local configurations underlie a number of well-studied graph parameters. For example, independent sets are precisely those sets all of whose vertices are self private neighbors, while sets in which every

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vertex has an external private neighbor are open irredundant sets. More generally, the combinations of external, internal and self private neighbors lead to the standard family of irredundance invariants, including the independence number $\alpha(G)$, the strong matching number, the upper irredundance number $IR(G)$, the upper open irredundance number $OIR(G)$, the open-open irredundance number $OOIR(G)$, the 1-dependence number $\alpha_1(G)$, and the closed-open irredundance number $COIR(G)$.

Several of these parameters have independent histories. Strong, or induced, matchings were studied by Cameron [3]. The concept of 1-dependence was introduced by Fink and Jacobson [11]. Open irredundance and related open variants were considered by Farley and Shachum [8]. The broader theory of irredundance was developed in many subsequent works, including Bollobas and Cockayne [2], Fellows, Fricke, Hedetniemi and Jacobs [9], Cockayne [5], Finbow [10], and the recent survey of Mynhardt and Roux [15]. The private neighbor viewpoint therefore provides a common language connecting domination, independence, induced matchings and irredundance.

Hedetniemi and Rall [13] recently introduced a family of graph invariants arising from private neighbors. Let G be a graph and let $U \subseteq V(G)$. A vertex is called a self, internal, or external private neighbor with respect to U according as it is isolated in $G[U]$, has degree one in $G[U]$, or lies outside U and has exactly one neighbor in U . The corresponding numbers of such vertices are denoted by $S(U)$, $I(U)$, and $E(U)$, respectively.

Instead of maximizing the size of a set U whose vertices themselves possess private neighbors of prescribed types, Hedetniemi and Rall considered the problem of maximizing the number of vertices of G which are private neighbors with respect to some set U . This gives rise to the following seven invariants.

- $SPN(G)$, the maximum number of self private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $SPN(G)$ -set.
- $IPN(G)$, the maximum number of internal private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $IPN(G)$ -set.
- $EPN(G)$, the maximum number of external private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $EPN(G)$ -set.
- $ESPN(G)$, the maximum number of external and self private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $ESPN(G)$ -set.
- $EIPN(G)$, the maximum number of external and internal private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $EIPN(G)$ -set.
- $ISPN(G)$, the maximum number of internal and self private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $ISPN(G)$ -set.
- $EISPN(G)$, the maximum number of external, internal, and self private neighbors with respect to a set $U \subseteq V(G)$. Such a set U is called an $EISPN(G)$ -set.

These invariants measure the largest possible number of vertices that can be forced into prescribed private neighbor roles. This shift from maximizing the certificate set U to maximizing the vertices certified by U leads to a new family of extremal and algorithmic questions.

The basic examples already show that the new parameters behave differently from the classical irredundance parameters. Hedetniemi and Rall [13] determined exact formulae for paths, cycles and complete bipartite graphs, and they computed values for several small grid graphs. They also proved the sharp tree lower bound $IPN(T) \geq n(T)/2$ for every nontrivial tree T . The sharpness examples are coronas of paths, which demonstrate that internal private neighbors alone can cover only half of the vertices in the worst case. This naturally raises the question of whether allowing external and self private neighbors gives a larger universal lower bound on trees.

For bipartite graphs, the trivial lower bound $ESPN(G) \geq \alpha(G) \geq n(G)/2$ follows by taking a larger bipartition class as U , since every vertex of an independent set is self private with respect to that set. In the class of trees this lower bound is far from best in all known examples. Hedetniemi and Rall [13] proposed that the optimal constant should be $4/5$. Their extremal candidate is the following family: start with a path P_k with spine vertices $x_1, \dots, x_k$, attach two leaves to each x_i , and also attach to each x_i a pendent path of length 2. The resulting tree has order $5k$, and it was observed that its $ESPN$ -value is $4k$. This led to the first problem addressed in the present paper.

Problem 1.1 (Hedetniemi–Rall [13]). *Find the largest positive constant c such that $ESPN(T) \geq cn(T)$ for every tree T .*

We resolve this problem by proving Theorem 1.2.

Theorem 1.2. *For every tree T , $ESPN(T) \geq 4|V(T)|/5$, and the constant $4/5$ is best possible.*

The other two questions considered here concern grid graphs. Grid graphs have long served as test objects for domination parameters, because their simple local structure often conceals delicate periodic behaviour. In particular, perfect domination, total perfect domination and efficient domination in grids have been studied by several authors. Cockayne, Hartnell, Hedetniemi and Laskar [4] introduced perfect domination. Bernhard, Hedetniemi and Jacobs [1] studied efficient sets, while Dunbar et al. [7] investigated nearly perfect sets. Klostermeyer and Goldwasser [14] characterized total perfect codes in grid graphs.

These efficiency notions are directly connected with the private neighbor parameters. If $EIPN(G) = n(G)$, then every vertex of G can be made either an external or an internal private neighbor with respect to some set U ; this is the private neighbor analogue of total efficient domination. Similarly, $EISP_N(G) = n(G)$ means that every vertex can be made external, internal or self private with respect to a single set U . Thus exact values of $EIPN$ and $EISP_N$ on grids connect the new parameters of Hedetniemi and Rall with the older literature on efficient domination and total perfect codes.

Let $P_n \square P_m$ denote the Cartesian product of two paths. Based on their computations for small grids, Hedetniemi and Rall [13] proposed the following two conjectures.

Conjecture 1.3 (Hedetniemi–Rall [13]). *For every integer $m \geq 2$, $EIPN(P_2 \square P_m) = 2m$.*

Conjecture 1.4 (Hedetniemi–Rall [13]). *For every integer $k \geq 1$, $EISPN(P_3 \square P_{2k}) = 6k$, and for every integer $m \geq 2$, $EISPN(P_4 \square P_m) = 4m$.*

The first conjecture asserts that every ladder graph $P_2 \square P_m$ is extremal for $EIPN$ in the strongest possible sense: all of its $2m$ vertices can be counted as external or internal private neighbors with respect to one suitable set. The second conjecture makes the analogous assertion for two further infinite grid families, but now allowing self private neighbors as well. Since no parameter of the form $E(U) + I(U) + S(U)$ can exceed $|V(G)|$, these conjectures are equivalent to constructing optimal sets U that certify all vertices of the corresponding grids. Our next two theorems prove both conjectures.

Theorem 1.5. *For every integer $m \geq 2$, $EIPN(P_2 \square P_m) = 2m$.*

Theorem 1.5 immediately determines the value of $EISPN$ on ladders, since $EIPN(G) \leq EISPN(G) \leq |V(G)|$ for every graph G .

Corollary 1.6. *For every integer $m \geq 2$, $EISPN(P_2 \square P_m) = 2m$.*

Theorem 1.7. *For every integer $k \geq 1$, $EISPN(P_3 \square P_{2k}) = 6k$. Moreover, for every integer $m \geq 2$, $EISPN(P_4 \square P_m) = 4m$.*

The rest of the paper is organized as follows. In Section 2 we introduce notation and record the elementary upper bound of the parameters. In Section 3 we prove the sharp tree bound for $ESPN$, including the lower bound for forests and the extremal construction showing that the constant $4/5$ cannot be improved. In Section 4 we prove the two grid graph conjectures by constructing suitable sets U for ladders and for the grid families $P_3 \square P_{2k}$ and $P_4 \square P_m$. We conclude with several remarks and open questions.

2 Notation and elementary bounds

2.1 Notation

All graphs in this paper are finite, simple and undirected. For a graph G , we write $V(G)$ and $E(G)$ for its vertex set and edge set, and set $n(G) := |V(G)|$. For $x \in V(G)$, let $N_G(x) := \{y \in V(G) : xy \in E(G)\}$ and $N_G[x] := N_G(x) \cup \{x\}$. If $U \subseteq V(G)$, write $p_U(x) := |N_G(x) \cap U|$. When G is clear from the context, we omit the subscript G .

A tree is a connected acyclic graph, and a forest is a graph whose components are trees. A leaf is a vertex of degree 1. A path with vertices $v_0, \dots, v_d$ in this order will be written as $v_0 v_1 \dots v_d$.

For two graphs G and H , their Cartesian product $G \square H$ is the graph with vertex set $V(G) \times V(H)$, where two vertices (g, h) and (g', h') are adjacent if and only if either $g = g'$ and $hh' \in E(H)$, or $h = h'$ and $gg' \in E(G)$. We denote by P_m the path on m vertices. For the grid graph $P_r \square P_m$, we write $v_{i,j} := (i, j)$, where $1 \leq i \leq r$ and $1 \leq j \leq m$. Thus $v_{i,j}$ is adjacent to $v_{i',j'}$ precisely when $|i - i'| + |j - j'| = 1$.

Definition 2.1 (Private neighbor parameters). Let G be a graph and let $U \subseteq V(G)$. Recall that $p_U(x) = |N_G(x) \cap U|$ for $x \in V(G)$. A vertex x is called:

- an *external private neighbor* with respect to U if $x \in V(G) \setminus U$ and $p_U(x) = 1$;
- an *internal private neighbor* with respect to U if $x \in U$ and $p_U(x) = 1$;
- a *self private neighbor* with respect to U if $x \in U$ and $p_U(x) = 0$.

Let $E_G(U)$, $I_G(U)$ and $S_G(U)$ denote respectively the numbers of external, internal and self private neighbors with respect to U . When the graph G is clear from the context, we simply write $E(U)$, $I(U)$ and $S(U)$.

Since $SPN(G) = \alpha(G)$ is classical, we focus on the following six parameters:

$$\begin{aligned}
IPN(G) &:= \max_{U \subseteq V(G)} I_G(U), & EPN(G) &:= \max_{U \subseteq V(G)} E_G(U), \\
ESPN(G) &:= \max_{U \subseteq V(G)} (E_G(U) + S_G(U)), & EIPN(G) &:= \max_{U \subseteq V(G)} (E_G(U) + I_G(U)), \\
ISP(N)(G) &:= \max_{U \subseteq V(G)} (I_G(U) + S_G(U)), & EISP(N)(G) &:= \max_{U \subseteq V(G)} (E_G(U) + I_G(U) + S_G(U)).
\end{aligned}$$

2.2 Elementary bounds

Proposition 2.2. *Let G be a graph. Then each of*

$$IPN(G), \quad EPN(G), \quad ESPN(G), \quad EIPN(G), \quad ISP(N)(G), \quad EISP(N)(G)$$

is at most $n(G)$. In particular, if for some $U \subseteq V(G)$ every vertex of G is counted by one of the types appearing in a given parameter, then that parameter is equal to $n(G)$.

Proof. Fix $U \subseteq V(G)$. A vertex $x \in V(G)$ can contribute to at most one of $E_G(U)$, $I_G(U)$, and $S_G(U)$. Indeed, membership in $E_G(U)$ requires $x \notin U$, while membership in $I_G(U)$ or $S_G(U)$ requires $x \in U$; moreover, $x \in U$ cannot contribute to both $I_G(U)$ and $S_G(U)$, since these require respectively $p_U(x) = 1$ and $p_U(x) = 0$. Hence $E_G(U) + I_G(U) + S_G(U) \leq n(G)$. All six upper bounds follow by deleting nonnegative terms and taking the maximum over all $U \subseteq V(G)$. The final assertion follows from this upper bound and the assumed construction of U . $\square$

3 External and self private neighbors in trees

We first prove the lower bound in the slightly more general setting of forests.

Lemma 3.1. *For every forest F , $ESPN(F) \geq \frac{4}{5}|V(F)|$.*

Proof. We argue by induction on $n = |V(F)|$. The empty forest is trivial. If F is disconnected, apply the induction hypothesis to each component and take the union of the corresponding sets; since different components have no edges between them, the numbers of counted vertices add. It remains to consider a tree T .

For $n \leq 4$, the assertion follows from explicit choices. If $n = 1$, take $U = V(T)$. If $n = 2$, take one endpoint of the unique edge. If $n = 3$, then $T = P_3$, and taking the middle vertex as U counts

all vertices. If $n = 4$, then T is either P_4 or $K_{1,3}$; for P_4 , take the two endpoints as U , while for $K_{1,3}$, take the center as U . In each case all vertices are counted.

Assume $n \geq 5$. If T has diameter at most 2, then T is a star, and taking its center as the only vertex of U counts every vertex. We may therefore choose a longest path $v_0v_1 \cdots v_d$ with $d \geq 3$. Let B be the vertex set of the component of $T - v_2v_3$ containing v_2 , and put $b = |B|$. By maximality of the path, every vertex of B has distance at most 2 from v_2 ; otherwise a vertex of B at distance at least 3 from v_2 would give a path to v_d longer than $v_0v_1 \cdots v_d$. Thus every neighbor of v_2 in B is either a leaf, or all its neighbors in B other than v_2 are leaves. Since $v_0, v_1, v_2 \in B$, we have $b \geq 3$. Consider the following three cases.

Case 1. $b \geq 5$.

By induction, choose $U' \subseteq V(T - B)$ counting at least $4(n - b)/5$ vertices of $T - B$. Define $U_B \subseteq B$ as follows. For each neighbor x of v_2 lying in B , put x into U_B , and put none of the leaves adjacent to x into U_B ; here, if x itself is a leaf, this simply puts x into U_B . Do not put v_2 into U_B . Then every vertex of $B \setminus \{v_2\}$ is counted: each neighbor of v_2 in B that is a leaf is self private, each neighbor x of v_2 in B that is not a leaf is self private, and every leaf adjacent to such a vertex x is external private. Since $v_2 \notin U' \cup U_B$, the status of every vertex in $T - B$ is unchanged. Therefore

$$E(U' \cup U_B) + S(U' \cup U_B) \geq \frac{4}{5}(n - b) + (b - 1) \geq \frac{4}{5}n.$$

Case 2. $b = 3$.

Then $B = \{v_0, v_1, v_2\}$. By induction, choose $U' \subseteq V(T - B)$ counting at least $4(n - 3)/5$ vertices of $T - B$. If $v_3 \notin U'$, set $U_B = \{v_1\}$; then v_1 is self private, and v_0, v_2 are external private through v_1 . If $v_3 \in U'$, set $U_B = \{v_0\}$; then v_0 is self private, v_1 is external private through v_0 , and v_2 is external private through v_3 . In both cases $v_2 \notin U_B$, so the status of all vertices in $T - B$ is unchanged. Hence at least $4(n - 3)/5 + 3 \geq 4n/5$ vertices are counted.

Case 3. $b = 4$.

Let $H = T - (B \cup \{v_3\})$. By induction, choose $U' \subseteq V(H)$ counting at least $4(n - 5)/5$ vertices of H . We keep $v_3 \notin U$. Since B has four vertices and contains the path $v_0v_1v_2$, exactly one of the following two configurations occurs. Either v_1 has a neighbor $w \neq v_0$ in B that is a leaf, or v_2 has a neighbor $w \neq v_1$ in B that is a leaf.

In the first case, set $U_B = \{v_1\}$. Then v_1 is self private, and v_0, w, v_2 are external private through v_1 , because $v_3 \notin U' \cup U_B$. In the second case, set $U_B = \{v_0, w\}$. Then v_0 and w are self private, v_1 is external private through v_0 , and v_2 is external private through w , again because $v_3 \notin U' \cup U_B$. No vertex of U_B is adjacent to a vertex of H , and $v_3 \notin U' \cup U_B$, so the status of every counted vertex of H remains unchanged. We count at least $4(n - 5)/5 + 4 = 4n/5$ vertices, completing the induction. $\square$

We now describe the extremal family showing that the constant $4/5$ cannot be improved. For $k \geq 1$, let T_k be the tree obtained from a path $x_1x_2 \cdots x_k$ by attaching to each x_i two leaves a_i, b_i and one pendent path $x_iy_iz_i$. Thus

$$V(T_k) = \{x_i, a_i, b_i, y_i, z_i : 1 \leq i \leq k\}$$

and $|V(T_k)| = 5k$.

Lemma 3.2. *For every $k \geq 1$, $ESPN(T_k) = 4k$.*

Proof. For the lower bound, take

$$U := \{a_i, b_i, y_i : 1 \leq i \leq k\}.$$

For every i , the vertices a_i, b_i, y_i belong to U and have no neighbor in U , so they are self private neighbors. The vertex z_i lies outside U and has the unique neighbor y_i in U , so it is an external private neighbor. The vertex x_i is not counted, since it has the three neighbors a_i, b_i, y_i in U . Hence $E(U) + S(U) = 4k$, and so $ESPN(T_k) \geq 4k$.

For the upper bound, fix an arbitrary $U \subseteq V(T_k)$. For each i , let

$$B_i := \{x_i, a_i, b_i, y_i, z_i\}.$$

We show that at most four vertices of B_i can be counted by $E(U) + S(U)$. Suppose instead that all five vertices of B_i are counted by $E(U) + S(U)$. Since a_i and b_i are leaves adjacent only to x_i , both are counted only if $a_i, b_i \in U$ when $x_i \notin U$, and only if $a_i, b_i \notin U$ when $x_i \in U$. If $x_i \notin U$, then x_i has at least two neighbors in U , namely a_i, b_i , and so x_i is not external private, a contradiction. Hence $x_i \in U$, and then $a_i, b_i \notin U$.

Since x_i itself is counted and lies in U , it must be self private, so none of its neighbors belongs to U ; in particular, $y_i \notin U$. Now z_i is a leaf adjacent only to y_i . Because $y_i \notin U$ and z_i is counted, we must have $z_i \in U$, making z_i self private. But then $y_i \notin U$ has two neighbors in U , namely x_i and z_i , and therefore y_i is not external private, again a contradiction. Thus at most four vertices of each block B_i are counted by $E(U) + S(U)$. The blocks $B_1, \dots, B_k$ partition $V(T_k)$, so $E(U) + S(U) \leq 4k$ for every U , and $ESPN(T_k) = 4k$ follows. $\square$

Proof of Theorem 1.2. Lemma 3.1 gives $ESPN(T) \geq 4|V(T)|/5$ for every tree T , so every $c \leq 4/5$ is valid. Lemma 3.2 gives an infinite family of trees T_k with $|V(T_k)| = 5k$ and $ESPN(T_k) = 4k = 4|V(T_k)|/5$. Therefore no constant larger than $4/5$ is valid for all trees. $\square$

4 Private neighbors in grid graphs

In this section we prove the conjectured exact values for the ladder graphs $P_2 \square P_m$ and for the two small-width grid families $P_3 \square P_{2k}$ and $P_4 \square P_m$. In each case we construct a set U such that every vertex of the grid is counted by the relevant parameter: by $EIPN$ for $P_2 \square P_m$, and by $EISP_N$ for $P_3 \square P_{2k}$ and $P_4 \square P_m$. The equality then follows from Proposition 2.2.

Proof of Theorem 1.5. Write $x_j = v_{1,j}$ and $y_j = v_{2,j}$ for $1 \leq j \leq m$. We construct a set U so that $p_U(z) = 1$ for every vertex z .

Choose a set $J \subseteq \{1, \dots, m\}$ by

$$J = \begin{cases} \{j : j \equiv 2 \pmod{3}\}, & m \equiv 0 \pmod{3}, \\ \{j : j \equiv 1 \pmod{3}\}, & m \equiv 1 \pmod{3}, \\ \{j : j \equiv 2 \pmod{3}\}, & m \equiv 2 \pmod{3}. \end{cases}$$

With the convention that $0, m+1 \notin J$, every $j \in \{1, \dots, m\}$ satisfies

$$|\{j-1, j, j+1\} \cap J| = 1.$$

Indeed, if $m \equiv 0 \pmod{3}$, then $J = \{2, 5, \dots, m-1\}$ and the triples centered at vertices of J are exactly $\{1, 2, 3\}, \{4, 5, 6\}, \dots, \{m-2, m-1, m\}$. If $m \equiv 2 \pmod{3}$, then $J = \{2, 5, \dots, m\}$ and the corresponding sets are $\{1, 2, 3\}, \{4, 5, 6\}, \dots, \{m-3, m-2, m-1\}, \{m-1, m\}$. If $m \equiv 1 \pmod{3}$, then $J = \{1, 4, \dots, m\}$ and the corresponding sets are $\{1, 2\}, \{3, 4, 5\}, \{6, 7, 8\}, \dots, \{m-1, m\}$. In all three cases these sets are pairwise disjoint and cover $\{1, \dots, m\}$.

Now set

$$U := \{x_j, y_j : j \in J\}.$$

If $j \in J$, then $x_j, y_j \in U$ and $j-1, j+1 \notin J$, so x_j has the unique neighbor y_j in U , and y_j has the unique neighbor x_j in U . Hence both are internal private neighbors. If $j \notin J$, then $x_j, y_j \notin U$, and exactly one of the existing indices $j-1, j+1$ belongs to J . Thus x_j has exactly one neighbor in U , namely the corresponding horizontal neighbor x_{j-1} or x_{j+1} , and similarly y_j has exactly one neighbor in U . Hence both are external private neighbors.

Therefore all $2m$ vertices are counted by $E(U) + I(U)$, so $EIPN(P_2 \square P_m) \geq 2m$. The general upper bound $E(U) + I(U) \leq |V(P_2 \square P_m)| = 2m$ gives equality. $\square$

Proof of Theorem 1.7. First consider $P_3 \square P_{2k}$. Put $m = 2k$, and define

$$U := \{v_{1,j} : j \equiv 2, 3 \pmod{4}\} \cup \{v_{3,j} : j \equiv 0, 1 \pmod{4}\}.$$

Let $a_j = 1$ if $v_{1,j} \in U$, and let $a_j = 0$ otherwise. Then

$$(a_1, a_2, \dots, a_m) = (0, 1, 1, 0, 0, 1, 1, 0, \dots)$$

truncated after the even index m . Put $a_0 = a_{m+1} = 0$. For every j , if $a_j = 0$, then $a_{j-1} + a_{j+1} = 1$, while if $a_j = 1$, then $a_{j-1} + a_{j+1} \leq 1$. This follows from the four interior local patterns $0, 0, 1$; $0, 1, 1$; $1, 1, 0$; $1, 0, 0$, and at $j = 1$ or $j = m$ the boundary convention $a_0 = a_{m+1} = 0$ gives the same conclusion. The bottom-row indicator c_j , defined by $c_j = 1$ if $v_{3,j} \in U$ and $c_j = 0$ otherwise, satisfies $c_j = 1 - a_j$ for $1 \leq j \leq m$, and with $c_0 = c_{m+1} = 0$ it satisfies the analogous property.

Every middle-row vertex $v_{2,j}$ is outside U , and exactly one of $v_{1,j}, v_{3,j}$ belongs to U , so $v_{2,j}$ is an external private neighbor. For a top-row vertex $v_{1,j}$, the only possible neighbors in U are $v_{1,j-1}$

and $v_{1,j+1}$. If $v_{1,j} \notin U$, then $a_j = 0$, so it has exactly one neighbor in U and is external private; if $v_{1,j} \in U$, then $a_j = 1$, so it has at most one neighbor in U and is self private or internal private. The same argument, using the sequence (c_j) , applies to every bottom-row vertex. Thus every vertex of $P_3 \square P_{2k}$ is counted by $E(U) + I(U) + S(U)$, and the upper bound gives $EISPN(P_3 \square P_{2k}) = 6k$.

It remains to handle $P_4 \square P_m$. We encode each column by one of three symbols A, B, O as follows:

- type A : $v_{1,j}, v_{4,j} \in U$ and $v_{2,j}, v_{3,j} \notin U$;
- type B : $v_{2,j}, v_{3,j} \in U$ and $v_{1,j}, v_{4,j} \notin U$;
- type O : no vertex in the column belongs to U .

Define a word W_m of length m over $\{A, B, O\}$ by

$$W_m = \begin{cases} A(OBOA)^q, & m = 4q + 1, \\ AA(OBOA)^q, & m = 4q + 2, \\ BOA(OBOA)^q, & m = 4q + 3, \\ BOAA(OBOA)^{q-1}, & m = 4q, q \geq 1. \end{cases}$$

The set U is obtained column by column according to these rules.

Add a formal boundary symbol O before the first letter and after the last letter of W_m . From the displayed forms, every occurrence of O in W_m is flanked by one A and one B , every occurrence of B is flanked by two O 's, and every occurrence of A is adjacent only to symbols in $\{A, O\}$ and to at most one further A . Thus the possible local patterns with center in W_m are

Table 1: Local patterns in W_m .

center symbol	possible neighboring symbols
A	AOA, OAA, AAO
B	OBO
O	AOB, BOA

If column j is of type A , then $v_{1,j}, v_{4,j} \in U$. Since an A -column is not adjacent to a B -column and is adjacent to at most one further A -column, each of $v_{1,j}$ and $v_{4,j}$ has at most one neighbor in U , and is therefore self private or internal private. The vertices $v_{2,j}, v_{3,j}$ lie outside U and have unique neighbors in U , namely $v_{1,j}$ and $v_{4,j}$, respectively.

If column j is of type B , then $v_{2,j}, v_{3,j} \in U$, and the neighboring columns are both of type O . Hence $v_{2,j}$ and $v_{3,j}$ have exactly one neighbor in U , namely each other. The vertices $v_{1,j}, v_{4,j}$ lie outside U and have unique neighbors in U , namely $v_{2,j}$ and $v_{3,j}$, respectively.

Finally, if column j is of type O , then no vertex in the column belongs to U , and its two neighboring symbols consist of one A and one B . Therefore $v_{1,j}, v_{4,j}$ each have exactly one neighbor in U , coming from the adjacent A -column, while $v_{2,j}, v_{3,j}$ each have exactly one neighbor in U ,

coming from the adjacent B -column. Thus all four vertices in an O -column are external private neighbors.

Consequently every vertex of $P_4 \square P_m$ is counted by $E(U) + I(U) + S(U)$. Since $|V(P_4 \square P_m)| = 4m$, the general upper bound gives $EISP_N(P_4 \square P_m) = 4m$. $\square$

5 Remarks

The results proved above suggest several natural directions for further study. One immediate problem is to extend the exact grid results to larger fixed widths.

Problem 5.1. *Determine the exact values of $EIP_N(P_r \square P_m)$ and $EISP_N(P_r \square P_m)$ for every fixed integer $r \geq 5$ and all $m \geq 2$.*

The constructions for widths 2, 3 and 4 are periodic in the long direction of the grid. It would be interesting to know whether similar periodic constructions remain optimal for every fixed width r .

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